

Chemical Reactions and Equations

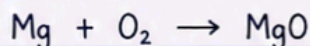
1. Chemical Reaction

- A process in which new substance is formed.

Signs of Chemical Reaction

- Change in colour
- Change in temperature
- Gas produced
- Change in state

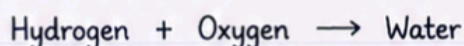
Example



2. Chemical Equation

- Representation of chemical reaction using symbols and formulae.

Word Equation



Chemical Equation



3. Balanced Chemical Equation

- Equation in which number of atoms on both sides are equal.

Why balancing is needed?

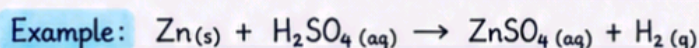
- Because of Law of Conservation of Mass.

Example



4. Important State Symbols

Symbol	Meaning
(s)	Solid
(l)	Liquid
(g)	Gas
(aq)	Aqueous solution

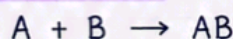


5. Types of Chemical Reactions

(A) Combination Reaction

- Two or more substances combine to form one product.

General Form



Example

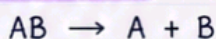


☑ Heat released →
Exothermic reaction

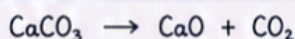
(B) Decomposition Reaction

- One substance breaks into simpler substances.

General Form



Example



Types

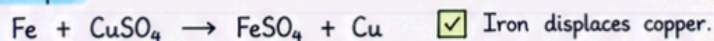
- Thermal decomposition → by heat
- Electrolytic decomposition → by electricity
- Photolytic decomposition → by sunlight



(C) Displacement Reaction

- More reactive element displaces less reactive element.

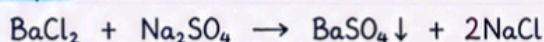
Example



(D) Double Displacement Reaction

- Exchange of ions between compounds.

Example



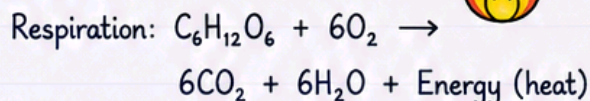
☑ White precipitate of BaSO_4 formed.

6. Exothermic & Endothermic Reactions

Exothermic Reaction

- Heat released.

Example



Endothermic Reaction

- Heat absorbed.

Example

Decomposition reactions.

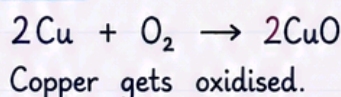


7. Oxidation and Reduction

Oxidation

- Gain of oxygen
- Loss of hydrogen

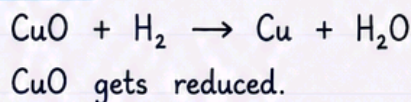
Example



Reduction

- Loss of oxygen
- Gain of hydrogen

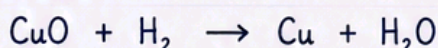
Example



8. Redox Reaction

- Oxidation and reduction occur together.

Example



- ★ H_2 oxidised
- ★ CuO reduced



9. Corrosion

- Slow eating away of metals by air/moisture.

Example

- Rusting of iron
- Green layer on copper
- Black layer on silver



Prevention

- Painting
- Oiling
- Galvanisation

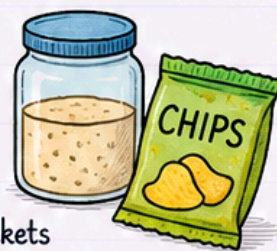


10. Rancidity

- Oxidation of fats and oils causing bad smell and taste.

Prevention

- Airtight containers
- Refrigeration
- Nitrogen gas in chips packets



Quick Tips

- ★ Balance the equation to obey the Law of Conservation of Mass.
- ★ State symbols show the physical state of substances.
- ★ Redox reactions are important in everyday life and industries.

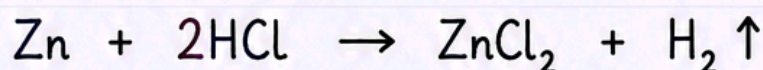


★ Most Important Reactions ★

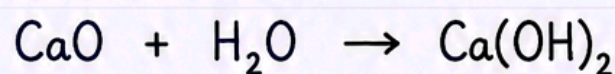
- Magnesium burning



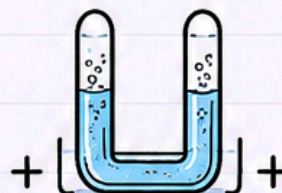
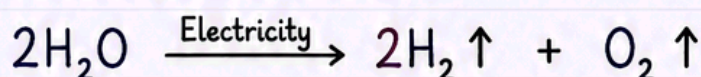
- Zinc with acid



- Whitewashing



- Electrolysis of water



⚡ One-Line Exam Revision

- Combination → many to one
- Decomposition → one to many
- Displacement → more reactive replaces less reactive
- Double displacement → ion exchange
- Oxidation → gain oxygen
- Reduction → lose oxygen
- Corrosion → metal damage
- Rancidity → bad smell in oils

